Systematic Map Codebook — D5.6

ILRI / Bristlepine Resilience Consultants — 2025–2026

Scope: Systematic map data extraction only (D4/D5). Not the systematic review (D8/D9).

Quick start for coders

Everything is on Google Drive — no GitHub or local setup needed.

→ Open the FT-R1a folder on Google Drive

Three steps:

1. Get your papers — open the `FT-R1a pdfs` subfolder in the Drive link above
2. Open the coding sheet — download `coding_ft_r1a_XX.csv`, save a copy as `coding_ft_r1a_INITIALS.csv` (e.g. `coding_ft_r1a_CS.csv`), and upload your completed file back to the same Drive folder
3. Code each paper using the schema below — one row per paper, all 18 fields — save regularly

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Rules while coding:

- Fill in `coder_id` with your initials — same initials every row
- Code independently — no discussion with the other coder until review
- If a field is genuinely not reported: leave blank or enter `not_reported`
- If unsure: make your best judgement and add a note in the `notes` column
- Do not look at the other coder's sheet

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Your sheet columns:

```
doi, filename,
publication_year, publication_type, country_region,
geographic_scale,
producer_type, marginalized_subpopulations, adaptation_focus,
domain_type,
```

`process_outcome_domains, indicators_measured, methodological_approach, purpose_of_assessment, data_sources, temporal_coverage, cost_data_reported, strengths_and_limitations, lessons_learned, validity_notes, coder_id, notes`

The first two columns (`doi`, `filename`) are pre-filled. Fill the rest.

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The 18 coding fields

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Use this as your reference while coding. Valid values must be used exactly as written (except free-text fields, marked ⚠️).

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1. `publication_year`

Year the study was published. Use the most recent version date if both preprint and journal exist. Valid values: integer, e.g. `2021`

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2. `publication_type`

Valid values: `journal_article` | `report` | `working_paper` | `thesis` | `other`

- `journal_article` — peer-reviewed journal (including open-access)
- `report` — project/programme report, evaluation report, policy brief
- `working_paper` — pre-publication working paper or discussion paper
- `thesis` — PhD or master's dissertation
- `other` — book chapter, conference paper, dataset paper

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3. `country_region`

Country or countries where the study was conducted. Use region label only if no countries are named. Valid values: free text, semicolons between multiple entries

Examples: `Tanzania` | `Ghana; Mali; Burkina Faso` | `East Africa` | `Global`

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4. `geographic_scale`

Valid values: `local` | `sub-national` | `national` | `multi-country` | `regional`

- `local` — village, community, single site
- `sub-national` — district, province, region within a country
- `national` — one country
- `multi-country` — two or more specific named countries
- `regional` — broad region without country-level focus

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5. `producer_type`

Select all that apply, semicolons between multiple values. Valid values: `crop` | `livestock` | `fisheries_aquaculture` | `agroforestry` | `mixed` | `undefined`

- `mixed` — explicitly mixed crop-livestock,
- `undefined` — generic "smallholder farmers" with no system specified

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6. `marginalized_subpopulations`

Code only what is explicitly stated — do not infer. Select all that apply. Valid values: `women` | `youth` | `people with disabilities` | `landless` | `indigenous_peoples` | `ethnic_minorities` | `migrant_seasonal_workers` | `other` | `none_reported`

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7. `adaptation_focus` ⚠️ *free text — emergent*

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The specific climate adaptation action, intervention, or practice the study tracks. 5–15 words.

Examples: `crop management practices` | `crop varieties and genetics` | `water management and irrigation` | `soil and land management` | `agroforestry` | `livestock management` | `livelihood diversification (non-farm)` | `financial strategies` | `climate information and advisory services` | `migration and mobility` | `social and community networking` | `post-harvest and consumption adjustments`

Deleted: `drought-tolerant maize varieties` | `seasonal climate forecasts for farmer decision-making` | `index-based livestock insurance` | `farmer field schools for climate-resilient agriculture`

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8. `domain_type`

Valid values: `adaptation_process` | `adaptation_outcome` | `both`

- Process — measures *how* farmers adopt, learn, participate (steps toward adaptation)
- Outcome — measures *what happened* (yields, income, wellbeing, resilience)
- Both — tracks both a process and its outcome in the same study
- Decision rule: adoption rate only → `adaptation_process`; yield impact only → `adaptation_outcome`

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9. `process_outcome_domains`

Select all that apply, semicolons between multiple values.

Code	Meaning
`knowledge_awareness_learning`	Awareness of climate change, knowledge of options, learning outcomes
`decision_making_planning`	Farmer decision-making, use of forecasts in planning
`uptake_adoption`	Rates of adoption of a practice or technology

`behavioral_change`	Changes in farming or land management behaviour
`participation_coproduction`	Participation in programmes, co-design, farmer group engagement
`institutional_governance`	Policy uptake, institutional change, governance
`access_information_services`	Access to extension, credit, markets, climate information
`yields_productivity`	Crop yield, livestock productivity, fish catch
`income_assets`	Household income, asset accumulation
`livelihoods`	Food security, livelihood diversification
`wellbeing`	Health, nutrition, subjective wellbeing
`risk_reduction`	Reduced exposure or sensitivity to climate shocks
`resilience_adaptive_capacity`	Adaptive capacity indices, resilience scores, vulnerability indices

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10. `indicators_measured` ⚠️ *free text*

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The specific indicators or metrics used. Extract actual indicators from methods/results. Include units.

Examples: `yield (t/ha), income (USD/season), food security score (HFIAS)` | `adoption rate (%), area under improved variety (ha)`

11. methodological_approach

Select all that apply. Valid values: `qualitative` | `quantitative` | `participatory` | `modeling_with_empirical_validation` | `experimental`

- `participatory` — only when participation IS the primary design (PRA, photovoice), not when farmers are merely surveyed
- `modeling_with_empirical_validation` — crop/climate/agent-based models validated with field data
- `experimental` — Includes Randomized Controlled Trials (RCTs), Natural Experiments, and Quasi-Experimental approaches. Purely lab and field experiments generally should not be included in study.

Select both qualitative and quantitative for explicitly mixed-methods approaches.

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12. purpose_of_assessment

Valid values: `project_learning` | `program_evaluation` | `donor_reporting` | `national_reporting` | `research`

- `research` — academic study with no programme affiliation
- `project_learning` — M&E embedded in an ongoing project
- `national_reporting` — contributing to NAPs, NDCs, national monitoring frameworks

13. data_sources

Select all that apply. Valid values: `surveys` | `administrative_data` | `remote_sensing` | `participatory_methods` | `secondary_data` | `other`

14. temporal_coverage

Valid values: `cross_sectional` | `seasonal` | `longitudinal`

- `cross_sectional` — single point in time or single season
- `seasonal` — intra-annual, across one or a few seasons
- `longitudinal` — multi-year, panel data, repeated measures
- `repeated_cross_sectional` — multi-year, repeated measures with different individuals/units

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15. `cost_data_reported`

Valid values: `yes` | `no` Code `yes` only if the paper explicitly reports cost figures or cost-effectiveness ratios. Qualitative mention of "limited resources" = `no`.

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16. `strengths_and_limitations` ⚠️ *free text*

Deleted: 16. `equity_inclusion`
Select all that apply. Use `none_reported` if equity is not mentioned. Valid values: `gender` | `youth` | `land_tenure` | `disability` | `other` | `none_reported`

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Author-reported strengths and limitations combined (protocol Table 3). Label each component.

Format: `Strength: <text>. Limitation: <text>.` Look in Discussion, Conclusion, and Limitations sections. Extract verbatim where possible.

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17. `lessons_learned` ⚠️ *free text*

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Key lessons or recommendations the authors report. 1–3 sentences. Focus on what they say should be done differently — not findings. Enter `not_reported` if absent.

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18. `validity_notes` ⚠️ *free text*

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Notes on methodological robustness that affect how much weight this study carries. Flag: small sample (n < 30), single site, self-reported outcomes, short follow-up, no control group, attrition not reported. Enter `none_flagged` if no concerns.

Reconciliation (Step 2)

After both coders finish independently:

1. Open both sheets side by side
2. For each paper and each field:
    - Agree → enter that value in `reconciled.csv`
    - Disagree → discuss until you reach agreement, then enter it
3. Record the reason for any substantive disagreement in `reconciliation_notes`
4. Cannot agree → escalate to lead researcher

Do not average numeric fields — always discuss to a single agreed value.

Commented [2]: @zarrar@bristlep.com cab you update this please?

Inter-rater reliability (IRR)

Commented [3]: Maybe we keep this for the LLM?

After reconciliation, run:

```
python scripts/step11_irr_analysis.py \
  --h1 documentation/coding/systematic-map/rounds/FT-R1a/coding_H1.csv \
  --h2 documentation/coding/systematic-map/rounds/FT-R1a/coding_H2.csv \
  --gold documentation/coding/systematic-map/rounds/FT-R1a/reconciled.csv \
  --llm documentation/coding/systematic-map/rounds/FT-R1a/coding_LLM.csv \
  --out documentation/coding/systematic-map/rounds/FT-R1a/irr_R1a.json
```

Kappa interpretation

κ	Interpretation	Action
< 0.20	Slight	Major revision — field may need redesign
0.21–0.40	Fair	Significant revision — add examples, tighten definitions

0.41–0.60	Moderate	Revision needed — review disagreement patterns
0.61–0.80	Substantial	Approaching threshold — fine-tuning only
> 0.80	Almost perfect	Ready to lock

Threshold to lock a closed field: κ ≥ 0.60 AND sensitivity ≥ 0.80 in the same batch.

Emergent field (`adaptation_focus`)

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Tracked by category saturation, not kappa. Saturated when no new codes appear across two consecutive batches.

Stopping rules

- Closed fields — locked once κ ≥ 0.60 AND sensitivity ≥ 0.80 in the same batch.

  Locked fields are not retested.
- Emergent fields — saturated when no new codes appear across two consecutive batches.
- Full corpus run — proceed to step15 once all closed fields are locked AND both emergent fields are saturated. Expected: 3–4 rounds (300–400 papers).

How rounds work

Round naming

Label	Who codes	Papers

`FT-R1a`	H1 + H2 + LLM independently	First 100 papers
`FT-R1b`	LLM only (re-codes same R1a papers after criteria update)	Same 100 papers as R1a
`FT-R2a`	H1 + H2 + LLM independently	Next 100 papers (101–200)
`FT-R2b`	LLM only (re-codes R2a papers after criteria update)	Same 100 as R2a

`a` rounds = full human + LLM pass on new papers. `b` rounds = LLM verification only, same papers, after criteria update. Human coders only ever code in `a` rounds.

Each round — six steps

```
Step 1 — Code independently
  H1, H2, and LLM all code the same 100 papers. No discussion.
  ↓
Step 2 — Reconcile (H1 + H2 only)
  Compare field by field. Agree on gold standard → reconciled.csv
  ↓
Step 3 — Compute IRR
  Run step11. Kappa + sensitivity per field and overall.
  ↓
Step 4 — Review disagreements
  Which fields underperformed? Was guidance unclear? Categories
missing?
  ↓
Step 5 — Update codebook and LLM criteria
  Update CODEBOOK.md with clarified definitions and new
examples.
  Copy llm-criteria/criteria_vN.yml → criteria_v(N+1).yml with field-
level
```

```
  improvements. Log the change in llm-criteria/CHANGELOG.md.
  ↓
Step 6 — Check stopping rules
  All closed fields locked AND emergent fields saturated → run full corpus.
  Otherwise → LLM re-codes same batch (FT-R1b) to verify improvement, then
  proceed to next round (FT-R2a) with updated criteria.
```

LLM criteria tracking

After each round's reconciliation, the LLM extraction prompt is updated and versioned — analogous to how the abstract screening criteria evolved across calibration rounds.

```
documentation/coding/systematic-map/llm-criteria/
├── criteria_v1.yml    ← loaded by step15; used in FT-R1a
├── criteria_v1.md     ← human-readable companion
├── criteria_v2.yml    ← updated after FT-R1a reconciliation; used in FT-R1b and FT-R2a
└── CHANGELOG.md       ← one line per version: round, date, what changed and why
```

To update after a round:

1. Review fields where LLM underperformed (κ < 0.60 or sensitivity < 0.80)
2. Copy `criteria_vN.yml` → `criteria_v(N+1).yml`
3. Add `r1_further_guidance:` (or `r2_`, etc.) to any field that needed clarification
4. Add one line to `CHANGELOG.md`
5. Update `step15_criteria_yml` in `config.py` to point to the new file

Folder structure

```
documentation/coding/systematic-map/
├── CODEBOOK.md              ← this document
├── llm-criteria/
│   ├── criteria_v1.yml      ← machine-readable LLM prompt (loaded by step15)
│   ├── criteria_v1.md       ← human-readable companion
│   └── CHANGELOG.md
└── rounds/
    ├── FT-R1a/
    │   ├── FT-R1a pdfs/         ← full-text PDFs/HTMLs (on Google Drive, not in git)
    │   ├── coding_ft_r1a_XX.csv     ← blank template (coders save-as
coding_ft_r1a_CS.csv etc.)
    │   ├── reconciled_ft_r1a.csv    ← gold standard after reconciliation
    │   └── irr_ft_r1a.json          ← kappa, sensitivity, field-level stats (generated)
    │   [on Drive only: coding_ft_r1a_CS.csv, coding_ft_r1a_JD.csv,
coding_ft_r1a_LLM.csv]
    ├── FT-R1b/                  ← LLM re-code only (same papers as R1a)
    ├── FT-R2a/              ← next 100 papers
    └── ...

documentation/coding/abstract-screening/
└── criteria/                ← eligibility criteria used in abstract screening (locked)
```

Round log

Commented [4]: Do we want to talk a bit about the random selection convergence process and number of rounds?

Round	Papers	Date	H-H κ	LLM sensitivity	Status

FT-R1a	100	—	—	—	⏳ In progress
FT-R1b	same 100	—	—	—	— Not started
FT-R2a	100	—	—	—	— Not started

Update after each round's IRR run.

Quick reference — valid values

Field	Values
`publication_type`	journal_article, report, working_paper, thesis, other
`country_region`	free text
`geographic_scale`	local, sub-national, national, multi-country, regional
`producer_type`	crop, livestock, fisheries_aquaculture, agroforestry, mixed, undefined
`marginalized_subpopulations`	women, youth, people with disabilities, landless, indigenous_peoples, ethnic_minorities, migrant_seasonal_workers, other, none_reported
`adaptation_focus`	free text

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domain_type	adaptation_process, adaptation_outcome, both
process_outcome_domains	knowledge_awareness_learning, decision_making_planning, uptake_adoption, behavioral_change, participation_coproduction, institutional_governance, access_information_services, yields_productivity, income_assets, livelihoods, wellbeing, risk_reduction, resilience_adaptive_capacity
indicators_measured	free text
methodological_approach	qualitative, quantitative, participatory, modeling_with_empirical_validation; experimental
purpose_of_assessment	project_learning, program_evaluation, donor_reporting, national_reporting, research
data_sources	surveys, administrative_data, remote_sensing, participatory_methods, secondary_data, other
temporal_coverage	cross_sectional, seasonal, longitudinal
cost_data_reported	yes, no
strengths_and_limitations	free text (Strength: ... / Limitation: ...)

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Deleted: equity_inclusion

Deleted: gender, youth, land_tenure, disability, other, none_reported

`lessons_learned`	free text
`validity_notes`	free text